

→ KV Cache

Per decoding step $O(N^2) \rightarrow O(N)$ but increases VRAM

- memory bandwidth vram $\rightarrow$ onchip memory

layer $N (QK^T)V$

- Smaller KV Cache saves memory transfer overhead also

$\Rightarrow$ MHA and GQA

Each attention layer

W_q	$D \times n_{heads} \cdot d_k$
W_k	$D \times n_{heads} \cdot d_k$
W_v	$D \times n_{heads} \cdot d_v$

$\downarrow$
input
dimension

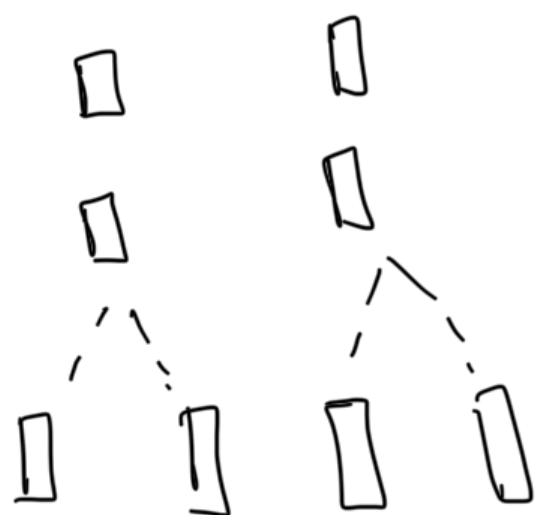
MHA

all query heads get the same key and value

MHA $\rightarrow$ MHA Reduces representational power of attn layer
 $\downarrow$ KV Cache memory + bandwidth performance

GQA

Values
Keys
Queries



larger models suffer relatively less from memory bw overhead from attention, as the KV-cache scales w/ model dimension while flops and parameters scale w/ the square of the model dimension

MQA ———— x GQA ———— MHA

T1 T2 T3 T4 [D dim]

Q1 Q2 Q3 Q4

d_{11} $-\infty$ $-\infty$ $-\infty$ k_1
 d_{21} d_{22} $-\infty$ $-\infty$ k_2
 d_{31} d_{32} d_{33} $-\infty$ k_3
 d_{41} d_{42} d_{43} d_{44} k_4

V1 V2 V3 V4

Z1 Z2 Z3 Z4

[D dimension]

01 02 03 04 05 06 07 08 09 10 11 12